

Test 4 Review

This is a review of the material covered after test 3 but will be on the final.

1 Line Integrals

Problem 1. Evaluate

$$\int_C 3x^2 - 2y \, ds$$

where C is the line segment from $(3, 6)$ to $(1, -1)$

Problem 2. Using the same C from the previous problem, compute

$$\int_C 3x^2 - 2y \, dx \text{ and } \int_C 3x^2 - 2y \, dy$$

Problem 3. Find area under the surface $f(x, y) = 2yx^2 - 4x$ bounded by the curve obtained by traveling counterclockwise around the lower half of the semicircle with radius 3.

Problem 4. Evaluate

$$\int_C 6x \, ds$$

where C is the parabola $y = x^2$ from $x = -1$ to $x = 2$.

Problem 5. If $F(x, y, z)$ is a conservative vector field then we know $F = \nabla f(x, y, z)$ for some surface $f(x, y, z)$. We call such $f(x, y, z)$ the potential function. Find the potential function for

$$F(x, y, z) = (2z^4 - 2y - y^3)\hat{i} + (z - 2x - 3xy^2)\hat{j} + (6 + y + 8xz^3)\hat{k}$$

2 Vector Fields

Problem 6. Find the divergence and curl for each of the following vector fields. Sketch the vector field to verify your answer.

1. $\frac{x}{y}\hat{i} + \frac{y}{x}\hat{j}$

2. $\frac{y}{x}\hat{i} + \frac{x}{y}\hat{j}$

3. $\frac{x}{\sqrt{x^2+y^2}}\hat{i} + \frac{y}{\sqrt{x^2+y^2}}\hat{j}$

4. $yz\hat{i} + xz\hat{j} + xy\hat{k}$

5. $xy\hat{i} + xz\hat{j} + xyz^2\hat{k}$

Problem 7. Which of the vector fields above are conservative? How do you know?

3 Vector Fields through Line Integrals

Problem 8. Find the work done along the line segment $(0,0,0)$ to $(1,2,3)$ through the vector field $\langle xy, x^2, yz \rangle$

Problem 9. Evaluate

$$\int_C F \cdot dr$$

where $F(x, y, z) = e^{2x}\hat{i} + z(y+1)\hat{j} + z^3\hat{k}$ and C is given by the vector function $\vec{r}(t) = t^3\hat{i} + (1-3t)\hat{j} + e^t\hat{k}$ from $0 \leq t \leq 2$.

Problem 10. Find the work done to travel from $(2, -1)$ to $(5, 0)$ through the vector field

$$F(x, y) = \left\langle \frac{y}{3\sqrt[3]{x^2}} + \frac{1}{x}, \sqrt[3]{x} + 2y \right\rangle$$

Problem 11. Evaluate

$$\oint_C yx^2 dx + x^2 dy$$

where C is the left half of $x^2 + y^2 = 25$ and the y -axis (assume counterclockwise)

Problem 12. Evaluate

$$\oint_C F \cdot dr$$

where F is the 4-dimensional conservative vector field $F(x, y, z, w) = \nabla f(x, y, z, w)$

Problem 13. Evaluate

$$\int_C x^2 dy - yz dz$$

where C is the line segment from $(4, -1, 2)$ to $(1, 7, -1)$.

Problem 14. Evaluate

$$\oint_C xy dx + y^{2024} dy$$

where C is the curve traveling along the triangle with vertices $(-1, 2), (3, 5), (3, 7)$.

Problem 15. Find the work required to travel along the curve parameterized by $\vec{r}(t) = \langle \sin t, 1, t \rangle$ through the vector field $F(x, y, z) = \langle x, (xyz)^{x^2+y^2+z^2}, ze^z \rangle$ from $t = 0$ to $t = 1$.